STUDYNEST | CLASS 11 PHYSICS

QUESTION PAPER 1

Chapter 9: Mechanical Properties of Fluids

Maximum Marks: 40 | Suggested time: 60 minutes

Instructions: Answer all questions. Show calculations and units. Original practice material, not an official examination.

Define pressure. [1]
SI unit of dynamic viscosity? [1]
Continuity relation for incompressible flow? [1]
Bernoulli relation? [1]
Surface tension unit? [1]
Pressure rise at depth h? [1]
Stokes drag? [1]
Excess pressure in a liquid drop? [1]

Water at 3m depth, rho=1000,g=10: gauge pressure? [2]
If pipe area halves, incompressible speed? [2]
Under Bernoulli assumptions, faster horizontal flow means pressure? [2]
Force from p=5000 Pa over area .02 m^2? [2]
Drop S=.06 N/m,r=.003 m: excess pressure? [2]
Why is a soap bubble pressure difference twice a drop's? [2]

Pipe area 4 cm^2 at 2 m/s narrows to 1 cm^2. New speed? [3]
Horizontal water speeds 2 and 6 m/s, rho=1000. Find p_1-p_2. [3]
Stokes drag for r=.5 mm,=1 Pa s,v=.02 m/s? [3]
Explain why pressure rises with depth. [3]

Areas 6 cm^2 and 2 cm^2; inlet speed 1.5 m/s. Find outlet speed. [4]
Soap bubble r=2mm,S=.03N/m: excess pressure? [4]

End of paper - check working, units, signs and significant figures.